

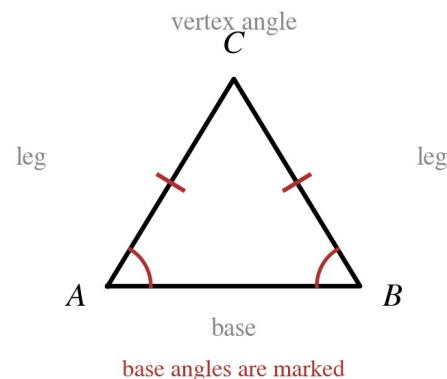
Speak Like A Mathematician:

legs (of an isosceles triangle):

vertex angle:

base:

base angles:



Isosceles and Equilateral Triangle Theorems

Base Angles Theorem: if two _____ of a triangle are congruent, then the _____ opposite them are congruent.

Converse of the Base Angles Theorem: if two _____ of a triangle are congruent, then the _____ opposite them are congruent.

Corollary: a triangle is equilateral if and only if it is _____, and each of its angles measures _____.

Example 1 - Name the Parts

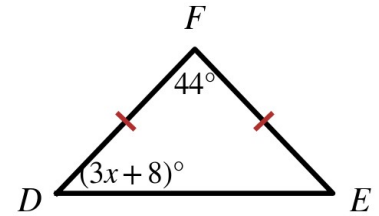
Use the isosceles triangle at the right.

- Name the legs, the base, and the vertex angle.
- Name the base angles. What does the Base Angles Theorem say about them?

Example 2 - Use the Base Angles Theorem

In $\triangle DEF$, the legs are marked congruent and the vertex angle measures 44° .

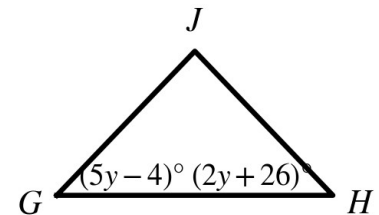
- Find the measure of each base angle.
- Write and solve an equation for x .



Example 3 - Use the Converse

In $\triangle GHJ$, the two base angles are marked with algebraic expressions.

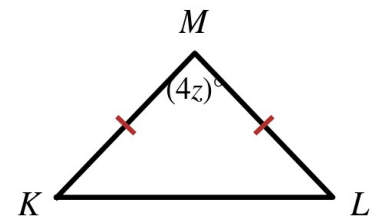
- Write and solve an equation for y .
- Find the measure of each base angle and of the vertex angle.
- What does the Converse let you conclude about the sides?



Example 4 - Equilateral and Equiangular

$\triangle KLM$ is equilateral.

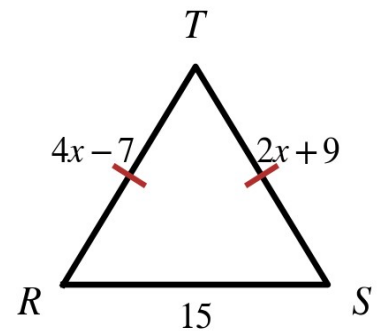
- Find z .
- Explain why an equilateral triangle must be equiangular.



Example 5 - Find x from the Sides

In $\triangle RST$, the two legs are marked congruent.

- Write and solve an equation for x .
- Find the length of each leg. Then find the perimeter.
- Could the base ever be longer than 50? Explain.



Example 6 - Two Isosceles Triangles

In the diagram, $\overline{AB} \cong \overline{AC}$, $\overline{BD} \cong \overline{BC}$, and $m\angle A = 38^\circ$. Find $m\angle DBC$.

- Find $m\angle ABC$ and $m\angle ACB$.
- Use $\triangle BDC$. What are $m\angle BDC$ and $m\angle BCD$?
- Find $m\angle DBC$.

